

Position, Velocity & Acceleration II

Numberbender | WORKSHEET

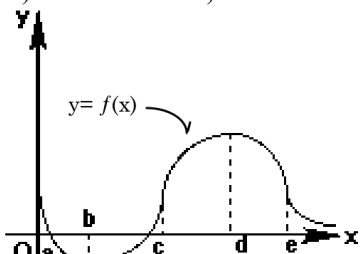


Name: _____ Date: _____ Score: _____

Analyze particle motion from position functions. Show all work.

CALCULUS I

Worksheet #73

In problems 1 – 4, the motion of a particle on a straight line is given by $s = t^3 - 6t^2 + 12t - 8$.	
1.	The distance s is increasing for A) $t < 2$ B) all t except $t = 2$ C) $1 < t < 3$ D) $t < 1$ or $t > 3$ E) $t > 2$
2.	The minimum value of the velocity occurs at $t =$ A) 1 B) 2 C) 3 D) 0 E) none of these
3.	The acceleration is positive A) when $t > 2$ B) for all t , $t \neq 2$ C) when $t < 2$ D) for $1 < t < 3$ E) for $1 < t < 2$
4.	The velocity of the particle is decreasing for A) $t > 2$ B) $t < 3$ C) all t D) $t < 1$ or $t > 2$ E) $t < 2$
5.	$\lim_{h \rightarrow 0} \frac{\cos\left(\frac{1}{2} + h\right)}{h}$ is: A) 1 B) nonexistent C) 0 D) -1 E) none
6.	On the graph of $y = f(x)$, $f'(x)$ and $f''(x)$ are both positive on which interval? A) $0 < x < a$ B) $b < x < c$ C) $c < x < d$ D) $d < x < e$ E) $x > e$ 
7.	The area in the first quadrant bounded by the curve $y = x^2$ and the line $y - x - 2 = 0$ is equal to A) $\frac{3}{2}$ B) $\frac{2}{3}$ C) $\frac{7}{6}$ D) $\frac{10}{3}$ E) $\frac{9}{2}$
8.	Let $f(x) = \begin{cases} \sqrt{x+4}-3; & x \neq 5 \\ c; & x = 5 \end{cases}$ and let f be continuous at $x = 5$. Then $c =$ A) $-\frac{1}{6}$ B) 0 C) $\frac{1}{6}$ D) 1 E) 6
9.	If $f(x) = x^5 + 2$, then its inverse function $f^{-1}(x)$ is A) $\frac{1}{x^5 + 2}$ B) $\sqrt[5]{x+2}$ C) $\sqrt[5]{x} + 2$ D) $\frac{1}{\sqrt[5]{x-2}}$ E) $\sqrt[5]{x-2}$
10.	A particle moves along a line so that at time t , where $0 \leq t \leq \pi$, its position is given by $s(t) = -4 \cos t - \frac{t^2}{2} + 10$. What is the velocity of the particle when its acceleration is zero? A) -5.19 B) 0.74 C) 1.32 D) 2.55 E) 8.13



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Position, Velocity & Acceleration II

Numberbender | ANSWER KEY



Answer key — for instructor use only.

Answers:

1. B	2. B	3. A	4. E	5. D	6. B	7. D	8. C	9. E	10. D	11. A
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